

The Speed of Influence: Sensory Bias and Interface Structure in Physical Theory

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March 26, 2026

Abstract

The constant c is properly understood as the universal propagation limit of causality, not a property specific to light. Its designation as the “speed of light” reflects a historical emphasis on electromagnetic measurement rather than a fundamental physical distinction. This paper examines how this emphasis has contributed to a sensory bias in the language, metaphors, and interpretive frameworks of physics.

Many foundational results can be derived independently of electromagnetic observation. The closure condition $C = g^2$ (Bisset 2026), derived from Newtonian force laws, expresses the self-confinement threshold of any field system in purely geometric terms without reference to electromagnetic quantities. Physical information may be more naturally understood as arising at the interfaces between interacting field systems rather than within any single field alone.

Biological systems offer a precedent for multi-field perception: avian cryptochrome proteins couple magnetic and electromagnetic fields within the visual system, producing orientation information that neither field generates alone. Humans possess related molecular hardware—CRY1 localized to retinal blue cone photoreceptors (Bartölke et al. 2025) and CRY2 correlated with navigation-related neural activation (Xu et al. 2021). Whether variant expression of these proteins corresponds to differences in perceptual experience remains an open and testable question.

1 The Speed of Influence

The constant c appears across multiple foundational theories, including Maxwell’s electrodynamics (1865), special relativity (Einstein 1905), and general relativity (Einstein 1915). While historically associated with light, its role is more general: it defines the maximum rate at which causal influence propagates. Massless excitations—electromagnetic radiation, gravitational waves (Abbott et al. 2016), gluon-mediated interactions—all saturate this limit. The universality of c is therefore a property of spacetime itself, not of any particular field. The naming convention persists due to the historical primacy of electromagnetic measurement, not because of an underlying physical restriction.

The closure condition $C = g^2$ (Bisset 2026) makes this independence explicit. Derived from two equations present in physics since 1687—the inverse square force law $F = k/r^2$ and the circular orbit condition $F = mv^2/r$ —the identity expresses the self-confinement threshold of any field system in purely geometric terms. No electromagnetic quantities appear. Applied at three known coupling constants, it recovers the proton charge radius ($g = 1/4$, 0.02% agreement), the Bohr radius and hydrogen ionisation energy ($g = \alpha$, nine significant figures), and the maximum stable compactness of neutron stars ($g = 1/\sqrt{2}$, 58/58 confirmed). A physicist operating entirely through pressure, acceleration, and tidal gradients could arrive at $C = g^2$ without detecting a single photon.

2 Sensory Bias in Physical Language

The language of physics reflects a strong visual bias. Terms such as “observer,” “observation,” “light cone,” and “spectral analysis” are rooted in visual or electromagnetic metaphors, despite the fact that the underlying mathematics does not require visual perception. Even the word “theory” derives from the Greek *theoria*—to look at, to contemplate.

This bias is historically understandable. Electromagnetic radiation provides a convenient measurement channel, and optical methods have dominated experimental practice. However, the reliance on visual metaphors may obscure the broader sensory-neutral structure of physical law. Einstein’s thought experiments, for example, often begin with what a light beam “looks like,” while the equivalence principle is fundamentally haptic: it concerns what a freely falling observer *feels*. A blind physicist would have framed the same principle through pressure and acceleration alone. The mathematics remains identical; only the sensory door changes.

3 Interface-Based Information

Physical systems are typically described in terms of fields or particles, with properties assigned within those domains. However, many observable phenomena arise most clearly at the boundaries between interacting systems rather than within any single field. Bisset’s closure framework localizes structure precisely at these boundaries: the strong closure surface at 0.841 fm, the electromagnetic closure surface at the Bohr radius, the gravitational closure surface at $R^* = 4GM/c^2$. In each case, the governing condition $C = g^2$ is satisfied at the interface where the field regime changes character, not in the interior.

This interfacial picture becomes especially clear when the gravitational container is locally removed. In the ISS water-sphere demonstration (Pettit & Sandlin 2016), a spherical water droplet in microgravity contains a small air bubble connected to a speaker. A 60 Hz pressure wave is transmitted through the air bubble into the surrounding water. Standing-wave patterns and surface geometries appear exclusively at the inner air-water interface—the boundary where the air bubble meets the water. The outer surface of the water sphere remains undisturbed, and the interior of the air bubble shows no structure. The observable geometry is generated only at the boundary between two media.

An analogy from sensory experience may clarify the claim. Flavor does not reside in any single receptor. It emerges at the intersection of gustatory, olfactory, thermal, and tactile signals. Remove any one channel and the experience changes fundamentally. The information lives in the combination—the interface—not in any individual input.

4 Sensory-Neutral Perspective

If physical laws are independent of sensory modality, then they should be expressible without reliance on any specific sensory framework. On Earth, every measurement occurs inside a permanent 1 *g* gravitational container—Einstein’s equivalence principle made literal. This container continuously damps weaker interfacial signals, potentially making boundary-dependent phenomena more difficult to resolve under terrestrial conditions. The ISS water-sphere demonstration illustrates this directly: remove the gravitational container and interfacial structure becomes visible. The information was always there. Gravity was masking it.

A sensory-neutral reconstruction of foundational physics would begin with interaction, propagation delay, local constraint, and feedback—elements already present in the formal structure of every major theory. The equivalence principle is haptic. The closure condition is geometric. The propagation limit c is causal. None of these require electromagnetic detection to state, derive, or verify. Reframing them in sensory-neutral terms does not change the physics. It may,

however, change which questions we think to ask—and which phenomena we consider worth investigating.

5 Biological Precedent

The claim that organisms can perceive information at field interfaces is not speculative. It is demonstrated.

Migratory birds possess cryptochrome proteins in their retinas that form radical pairs sensitive to the orientation of Earth’s magnetic field (Wiltschko & Wiltschko 2019). These proteins couple magnetic field information to the electromagnetic visual system, producing an orientation signal—experienced as a texture or shading overlaid on vision—that neither field generates alone. This is multi-field perception operating at a biological interface.

Humans possess related molecular hardware. CRY1 is expressed in the outer segments of short-wavelength (blue) cone photoreceptors in humans and great apes, suggesting a non-clock function (Bartölke et al. 2025). CRY2 expression in the human brain correlates with navigation-related neural activation, with individuals showing stronger circadian regularity also demonstrating better spatial orientation (Xu et al. 2021). The CRY1 R263Q variant has evolved independently at least five times in subterranean animal lineages, suggesting that cryptochrome function is subject to selection pressure when the sensory environment changes (Swaminathan et al. 2025).

Weak but measurable human magnetoreception has been demonstrated behaviorally—unconscious orientation to geomagnetic north under controlled conditions (Chae et al. 2022) and subconscious modulation of decision-making by magnetic field orientation, with sensitivity influenced by fasting state (Oh et al. 2025).

The sensory hardware exists. It is weakly expressed. It may be actively suppressed by the brain’s processing allocation: cross-modal competition, not integration, is the default mode when visual and non-visual signals arrive simultaneously (Yu et al. 2019; Mao et al. 2011). When the dominant sensory channel is weakened—through fatigue, fever, or altered neurochemistry—suppressed channels may compete more effectively for cortical processing, potentially producing perceptual artifacts that are currently interpreted as hallucination rather than as degraded rendering of normally sub-threshold input. Whether variant expression of cryptochrome proteins corresponds to differences in perceptual experience remains an open and testable question.

6 Discussion and Conclusion

The identification of c as the speed of influence rather than the speed of light is not a semantic preference. It reflects a structural observation: the constant c governs the propagation of all massless excitations, appears in field equations that have nothing to do with light, and can be derived from geometric closure conditions that reference no electromagnetic quantities.

Physics has historically privileged the electromagnetic channel—in its measurement instruments, its language, and its interpretive metaphors. This privileging is understandable: electromagnetic radiation is the most accessible and versatile observational tool available to sighted organisms on a planet bathed in light. But accessibility is not fundamentality. The equations do not require eyes. The equivalence principle is felt, not seen. The closure condition is geometric, not optical.

Biological systems demonstrate that multi-field perception is physically possible. Birds already sense at field interfaces. Humans possess the molecular hardware for similar perception, weakly expressed and largely unstudied. Whether this hardware contributes to perceptual experiences that current frameworks attribute to pathology or imagination is a question that biology, not psychology, is equipped to answer.

This paper does not claim that unnamed fields are real, that anomalous perception is validated, or that physics is incomplete. It claims something narrower: that a discipline built around one sensory channel may have systematically deprioritized questions about phenomena occurring outside that channel. Reframing foundational concepts in sensory-neutral terms does not change the physics. It may change which questions we think to ask.

What would Einstein have written if he was blind? Probably the same equations. But he might have asked different questions. And different questions lead to different discoveries.

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